**Appendix 3. Adapted QUADAS-2: signalling questions, judgment rules and study-level ratings**

Deep Learning for Inter-Patient ECG Arrhythmia Classification on the MIT-BIH Database: A Systematic Review

**1. Purpose**

Risk of bias was assessed with QUADAS-2, whose four domains were kept and whose signalling questions were rewritten for heartbeat classification on the MIT-BIH DS1/DS2 benchmark, with particular attention to data leakage. This adaptation has not been validated as a separate instrument. Each signalling question is answered Yes, No or Unclear (not reported), and each answer is recorded with the passage of the article that supports it. Domains 1 and 2 were defined in advance as critical. Ratings were made by the first author.

**2. Signalling questions and judgment rules**

Domain 1 - Record selection (critical)

Q1. Are the training and test records listed, or explicitly stated to follow DS1/DS2?

Q2. Is the split record-disjoint, with no beat or segment of a test record used for training (apart from the known 201/202 overlap)?

Q3. Were the four paced records (102, 104, 107, 217) excluded?

Rule: High if Q2 = No. Low if Q1 and Q2 = Yes. Unclear otherwise. Q3 = No raises an applicability concern, not a risk of bias.

Domain 2 - Index test, the model (critical)

Q1. Were architecture, hyperparameters and the number of training epochs chosen without DS2 (for example, with a validation set drawn from DS1)?

Q2. Were normalization, resampling (such as SMOTE) and data augmentation applied to training data only?

Q3. Were decision thresholds and class weights fixed without looking at DS2 results?

Q4. If the model adapts to test patients, does it use unlabelled data only? (Not applicable when there is no adaptation.)

Rule: High if any answer = No. Low if all applicable answers = Yes. Unclear if at least one answer is Unclear and none is No. Reporting the best epoch or best run on DS2 counts as No for Q1.

Domain 3 - Reference standard

Q1. Were the MIT-BIH cardiologist annotations used as the reference?

Q2. Was the mapping of MIT-BIH beat labels to AAMI classes reported and consistent with ANSI/AAMI EC57?

Rule: Low if both = Yes. High if a non-standard mapping moves beat types into or out of the S or V classes. Unclear if the mapping is not reported. Using three or four classes instead of five raises an applicability concern only.

Domain 4 - Flow and analysis

Q1. Were all annotated beats of the test records evaluated, or were any excluded beats counted and justified?

Q2. Do the reported results come from a single final evaluation on DS2, rather than the best of several test runs?

Q3. When a confusion matrix is given, are the reported metrics consistent with it?

Rule: High if Q2 = No, or if more than 5% of test beats were excluded without justification. Low if all answers = Yes. Unclear otherwise.

Overall rating. No overall score is computed. For the sensitivity analysis, a study is considered at low risk when Domains 1 and 2 are both rated Low.

**3. Summary of ratings (31 included studies)**

| Domain | Low | Unclear | High |
|---|---|---|---|
| D1 Record selection (critical) | 29 | 2 | 0 |
| D2 Index test (critical) | 3 | 22 | 6 |
| D3 Reference standard | 31 | 0 | 0 |
| D4 Flow and analysis | 8 | 18 | 5 |

## 4. Study-level ratings

| Study [ref] | D1 Record selection | D2 Index test | D3 Reference standard | D4 Flow and analysis |
|---|---|---|---|---|
| CNN-based models | | | | |
| Xiao H 2022 [20] | Low | Unclear - Validation curves shown without stating where the validation set came from, hyperparameter choice not described | Low | Unclear - A few test beats missing without explanation, single evaluation not stated |
| Xie 2023 [21] | Low | Unclear - Choice of hyperparameters and number of epochs not described | Low | Low |
| Farag 2023 [22] | Low | High - Final model chosen as the best variant on DS2 | Low | Low |
| Zhou F 2025 [23] | Low | Unclear - Fixed hyperparameters, no selection procedure described, oversampling limited to training data | Low | Low |
| Zhang J 2021 [24] | Low | High - Adversarial weight tuned and compared on DS2 | Low | Unclear - Some printed totals of test beats are incorrect |
| Merdjanovska 2021 [25] | Low | High - Number of epochs and best settings chosen on DS2 | Low | Unclear - No confusion matrix, overall accuracy not clearly defined |
| Berrahou 2026 [26] | Low | High - Final model variant selected on DS2 | Low | Unclear - No confusion matrix |
| Zubair 2024 [27] | Low | Unclear - No validation set or stopping criterion described | Low | Unclear - No test beat counts or confusion matrix, reported metrics do not agree with each other |
| Li Z 2025 [28] | Low | Unclear - Loss weight tunable, no validation set described | Low | Unclear - Results in the abstract differ from those in the tables, no confusion matrix |
| Cai 2021 [29] | Low | Low | Low | Low |
| Wang T 2020 [30] | Low | Unclear - Regularization tuned by trial and error, no validation described | Low | Low |
| Xu P 2022 [31] | Low | Unclear - Parameter tuning not detailed | Low | High - Results presented as the best of several runs |
| CNN-RNN hybrids | | | | |
| Chen A 2020 [32] | Low | Unclear - Validation curves without stated origin, rebalancing not described | Low | Unclear - S beat counts differ between tables, S results cannot be checked against the confusion matrix |
| Tao L 2021 [33] | Low | Unclear - Hyperparameters taken from an earlier intra-patient setup, tuning procedure for the inter-patient setting not described | Low | Low |
| Ran 2024 [34] | Low | Unclear - Loss weights chosen by a parameter sweep, data used not stated | Low | Unclear - About 374 N test beats missing without explanation, F class removed from the test |
| Wang Y 2023 [35] | Low | Unclear - Loss weights chosen by random search, tuning data not stated | Low | High - About 4,900 N test beats (about 10% of DS2) missing without explanation |
| Midani 2024 [36] | Low | Unclear - No validation described | Low | Unclear - No test beat counts or confusion matrix, decoding on DS2 not described |
| Essa 2021 [37] | Low | Unclear - Sampling ratio and input length apparently chosen on test results | Low | Unclear - Number of S test beats differs from DS2 without explanation |
| Lyu 2023 [38] | Low | Unclear - No validation described | Low | Unclear - Decoding on DS2 not described, near-perfect results |
| Attention-based models | | | | |
| Berrahou 2025 [39] | Low | Unclear - No validation set, training curves shown on the test set without stating whether they were used for stopping | Low | Unclear - No confusion matrix |
| Tao Y 2026 [40] | Unclear - Feature-enhancement block pre-trained on "knowledge data extracted from the ECG database", exclusion of DS2 not stated | Unclear - No validation described | Low | High - Only about 53% of DS2 beats tested, without explanation |
| Other architectures | | | | |
| Siouda 2021 [41] | Low | High - Number of features selected on DS2 results | Low | Unclear - Total number of test beats reported incorrectly, no confusion matrix |
| Zhou C 2023 [42] | Low | Unclear - No validation set, stopping and hyperparameter choices not described | Low | Unclear - No test beat counts and no confusion matrix |
| Zahid 2023 [43] | Low | Low | Low | Low |
| Qiu 2021 [44] | Unclear - Backbone pre-trained on beats of the "experimental" database, not stated whether limited to DS1 in the inter-patient setting | Unclear - No validation set described for the inter-patient setting | Low | High - Only about 38% of DS2 beats tested, without explanation |

| | | | | |
|---|---|---|---|---|
| Han 2021 [45] | Low | Unclear - Tuning and stopping criterion without a described validation set | Low | Unclear - About 6% of DS2 beats removed as noisy (reason given, but the removal method is not described) |
| Zou 2022 [46] | Low | Low | Low | Low |
| Models with test-patient adaptation (analysed separately) | | | | |
| Chen M 2020 [47] | Low | Unclear - Hyperparameter tuning not reported, unclear whether the first 5 minutes used for adaptation were also evaluated | Low | Unclear - No test beat counts and no confusion matrix |
| Chang 2026 [48] | Low | High - Clustering and segmentation settings chosen on DS2 results | Low | High - Results are the best performance on DS2 across settings |
| Wang G 2021 [49] | Low | Unclear - Last checkpoint used (no selection on DS2), but hyperparameter tuning not described | Low | Unclear - No test beat counts and no confusion matrix |
| Niu 2020 [50] | Low | Unclear - Selection of the “optimal model” not described, training curves shown on the test set | Low | Unclear - Small differences between two results tables |

Justifications are given for Unclear and High ratings. Reference numbers [20] to [50] follow the main article.